Exercises week 5 – Answers

8CM10 - Introduction to Quantum Physics and Structure of Matter

One-Dimensional Atom

1. By substituting the wave function $\psi(x) = Axe^{-bx}$ into:
$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \frac{e^2}{4\pi\varepsilon_0 x}\psi(x) = E\psi(x)$$
show that a solution can be obtained only for $b = \frac{1}{a0}$, and find the ground-state energy.

$\psi(x) = Axe^{-bx}$ gives $d\psi/dx = Ae^{-bx} - bAxe^{-bx}$ and $d^2\psi/dx^2 = -2bAe^{-bx} + b^2Axe^{-bx}$.
Then substituting into Equation 7.2 we have

$$-\frac{\hbar^2}{2m}(-2Abe^{-bx} + b^2Axe^{-bx}) - \frac{e^2}{4\pi\varepsilon_0 x}Axe^{-bx} = EAxe^{-bx}$$

Canceling common factors gives

$$\frac{\hbar^2 b}{m} - \frac{\hbar^2 b^2}{2m}x - \frac{e^2}{4\pi\varepsilon_0} = Ex \quad \text{or} \quad \left(\frac{\hbar^2 b}{m} - \frac{e^2}{4\pi\varepsilon_0}\right) + x\left(-\frac{\hbar^2 b^2}{2m} - E\right) = 0$$

For this expression to equal zero for all x, both terms in parentheses must be zero. Thus

$$\frac{\hbar^2 b}{m} = \frac{e^2}{4\pi\varepsilon_0} \quad \text{or} \quad b = \frac{me^2}{4\pi\varepsilon_0\hbar^2} = \frac{1}{a_0} \qquad \text{and} \qquad E = -\frac{\hbar^2 b^2}{2m} = -\frac{me^4}{32\pi^2\varepsilon_0^2\hbar^2}$$

2. Show that the probability density for the ground-state solution of the one-dimensional Coulomb potential energy has its maximum at $x = a_0$.

The probability density is $P(x) = |\psi(x)|^2 = A^2x^2e^{-2bx}$. To find the maximum, we set the first derivative equal to zero:

$$\frac{dP}{dx} = 2A^2xe^{-2bx} - 2bA^2x^2e^{-2bx} = 0$$

This has solutions at $x = 0$, $x = \infty$, and $x = 1/b = a_0$. The first two give minima and the third gives the maximum.

Hydrogen Atom

3. List the 16 possible sets of quantum numbers n, l, m_l of the $n = 4$ level of hydrogen

$l = 0$: $(4, 0, 0)$

$l = 1$: $(4, 1, +1), (4, 1, 0), (4, 1, -1)$

$l = 2$: $(4, 2, +2), (4, 2, +1), (4, 2, 0), (4, 2, -1), (4, 2, -2)$

$l = 3$: $(4, 3, +3), (4, 3, +2), (4, 3, +1), (4, 3, 0), (4, 3, -1), (4, 3, -2), (4, 3, -3)$

4. The $1s$ ($n = 1, l = 0$) wave function of the Schrödinger equation for the hydrogen atom is:

$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi}} a_0^{-3/2} e^{-r/a_0}$$

   a) Show that the corresponding probability density has its maximum value at $r = a_0$.
   b) What is the mean value of r for the $1s$ state of hydrogen? Use spherical coordinates for the integration overall space die to the spherically symmetric nature of the problem. What extra factor do you get integrating over a sphere due to the determination of the Jacobian?
   Hint:

$$\int_0^\infty x^n e^{-x} dx = n!$$

Take home message: When we label atoms according to their electron configurations, we are actually assigning wave functions to different electrons of that atom. Different states have different radial (and angular) densities. Keep this assignment of wavefunctions to electrons in mind when thinking of electron configurations.

a) Think twice before calculating numbers! There are two pitfalls easily overseen in this case:

- Positions at a certain radius lie on a spherical shell with radius $4\pi r^2$, implying that larges radius occupies more 'Area'. This needs to be represented in the integral.
- The wave function is real and therefore the probability is proportional to the square of the radial wave function

The radial probability density corresponding to $R_{1s}(r)$ is

$$P_{1s}(r) = 4\pi r^2 \psi_{1s}^2(r) = \frac{4r^2}{a_0^3} e^{-2r/a_0} \ .$$

The maximum of $P_{1s}(r)$ is found by setting the derivative to 0:

$$\frac{dP_{1s}(r)}{dr} = \frac{8r}{a_0^3} e^{-2r/a_0} - \frac{2}{a_0}\frac{4r^2}{a_0^3} e^{-2r/a_0} = 0 \ ,$$

which gives

$$\frac{8r}{a_0^3}\left(1 - \frac{r}{a_0}\right) e^{-2r/a_0} = 0 \ ,$$

with solution $r = a_0$.

b) The average radius value follows from

$$\begin{aligned}\langle r \rangle &= \iiint_V \underbrace{\psi_{1s}^*(x,y,z)\sqrt{x^2+y^2+z^2}\psi_{1s}(x,y,z)}_{\text{Expected value}}\, dxdydz \\ &= \iiint_V \underbrace{\psi_{1s}^*(r)r\psi_{1s}(r)}_{\text{Expected value}} \underbrace{r^2\sin(\theta)}_{|\text{Det J}|}\, drd\theta d\phi = \underbrace{\int_0^{2\pi}\int_0^{\pi}\sin(\theta)d\theta d\phi}_{=4\pi}\frac{1}{\pi}\int_0^{\infty}\frac{r^3}{a_0^3}e^{-2r/a_0}dr \ .\end{aligned}$$

Define $x = \dfrac{2r}{a_0} \longrightarrow \dfrac{dx}{dr} = \dfrac{2}{a_0} \longrightarrow dr = \dfrac{a_0}{2}dx \longrightarrow r = \dfrac{xa_0}{2}$ yields:

$$r_{\text{av}} = \frac{a_0}{2}\left(\frac{a_0}{2}\right)^3 \frac{4}{a_0^3}\int_0^{\infty} x^3 e^{-x}dx = \frac{a_0}{4}6 = \frac{3}{2}a_0 \ ,$$

where we have used that $\int_0^{\infty} x^n e^{-x}dx = n!$ for $n = 0, 1, 2, \ldots$ which may be shown by successive integration by parts, where the polynomial term gets reduced each time successively. For curtosy we show one reduction here:

$$\int_0^{\infty} x^3 \exp(-x)\, dx = \underbrace{-x^3\exp(-x)}_{\int \exp(-x)}\Big]_0^{\infty} - \underbrace{\int_0^{\infty} -3x^2\exp(-x)\, dx}_{\int \exp(-x),\quad \frac{d}{dx}x^3} = \qquad 3\int_0^{\infty} x^2\exp(-x)\, dx$$

Further reductions are done in the same way and result in similar multiplication with the power of the polynomial.

5. During class it was discussed that the Schrödinger equation for the hydrogen atom was separable into three simpler equations for each of the coordinates, which means that $\psi = R(r)\Theta(\theta)\Phi(\phi)$. These equations were:

   a) Radial Equation: $\frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}\left(-\frac{e^2}{4\pi\epsilon_0 r} - E_n\right) = l(l+1)$

   b) Azimuthal Equation: $\frac{1}{\Theta(\theta)}\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right) + l(l+1)\sin^2\theta = m_l^2$

   c) Polar Equation: $\frac{1}{\Phi(\phi)}\frac{d^2\Phi(\phi)}{d\phi^2} = -m_l^2$

   The $2p$ $(n = 2, l = 1)$ with $m_l = 1$ wave function and energy for hydrogen atoms are:

   $\psi_{211} = \frac{1}{8\sqrt{\pi}(a_0)^{\frac{5}{2}}} r\exp\left(-\frac{r}{2a_0}\right)\sin\theta\exp(i\phi)$, $\quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}$, $\quad E_2 = \frac{1}{4}\frac{-m}{2\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2$

   a) Identify the three different functions $R(r), \Theta(\theta), \Phi(\phi)$ ignoring any normalization constants.

   b) Show that ψ_{211} solves the Schrödinger equation by showing that $R(r), \Theta(\theta), \Phi(\phi)$ obey the different equations for the given l and m_l.
   Hint: remember that any normalization constant does not affect the validity of the Schrödinger equation.

a) Just pick out the explicit dependencies:

- $R(r) = r\exp\left(\frac{-r}{2a_0}\right)$
- $\Theta(\theta) = \sin(\theta)$
- $\Phi(\phi) = \exp(i\phi)$

b) For $R(r)$:

$$\begin{aligned}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) &= \frac{d}{dr}\left(r^2\left(\exp\left(\frac{-r}{2a_0}\right) - \frac{r}{2a_0}\exp\left(\frac{-r}{2a_0}\right)\right)\right) = \frac{d}{dr}r^2\exp\left(\frac{-r}{2a_0}\right) - \frac{d}{dr}\frac{r^3}{2a_0}\exp\left(\frac{-r}{2a_0}\right) \\ &= 2r\exp\left(\frac{-r}{2a_0}\right) - \frac{r^2}{2a_0}\exp\left(\frac{-r}{2a_0}\right) - \frac{3r^2}{2a_0}\exp\left(\frac{-r}{2a_0}\right) + \frac{r^3}{(2a_0)^2}\exp\left(\frac{-r}{2a_0}\right) \\ &= r\exp\left(\frac{-r}{2a_0}\right)\left(2 - \frac{4r}{2a_0} + \frac{r^2}{(2a_0)^2}\right) \\ \Rightarrow \frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) &= \left(2 - \frac{2r}{a_0} + \frac{r^2}{4a_0^2}\right)\end{aligned}$$

We now rewrite the effective potenital term in terms of a_0

$$\begin{aligned}-\frac{2mr^2}{\hbar^2}\left(\frac{-e^2}{4\pi\epsilon_0 r} - E_2\right) &= \frac{2rme^2}{\hbar^2 4\pi\epsilon_0} + \frac{2mr^2}{\hbar^2}\frac{-m}{8\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2 = \frac{2r}{a_0} - \frac{2r^2}{8}\left(\frac{me^2}{4\pi\epsilon_0\hbar^2}\right)^2 \\ &= \frac{2r}{a_0} - \frac{1}{4}\frac{r^2}{a_0^2}\end{aligned}$$

And finally adding all relevant terms:

$$\left(2 - \frac{2r}{a_0} + \frac{r^2}{4a_0^2}\right) + \frac{2r}{a_0} - \frac{1}{4}\frac{r^2}{a_0^2} = 2 = l(l+1), \text{ since } l = 1 \qquad \square$$

For $\Theta(\theta)$:

$$\begin{aligned}&\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right) = \frac{d}{d\theta}(\sin\theta\cos\theta) = \cos^2\theta - \sin^2\theta \Longrightarrow \\ &\frac{\sin\theta}{\sin\theta}(\cos^2\theta - \sin^2\theta) + 2\sin^2\theta = \cos^2\theta + \sin^2\theta = 1 = m_l^2, \text{ since } m_l = 1 \qquad \square\end{aligned}$$

For $\Phi(\phi)$:

$$\frac{d^2\Phi}{d\phi^2} = -\exp(i\phi) \Longrightarrow \frac{-\exp(i\phi)}{\exp(i\phi)} = -1 = -m_l^2, \text{ since } m_l = 1 \qquad \square$$

6. Show that the (1, 0, 0) and (2, 0, 0) wave functions listed in the table below are properly normalized. *Hint:* $\int_0^\infty x^n e^{-ax}dx = \frac{n!}{a^{n+1}}$

TABLE 7.1 Some Hydrogen Atom Wave Functions

n	l	m_l	$R(r)$	$\Theta(\theta)$	$\Phi(\phi)$
1	0	0	$\frac{2}{a_0^{3/2}}e^{-r/a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
2	0	0	$\frac{1}{(2a_0)^{3/2}}\left(2 - \frac{r}{a_0}\right)e^{-r/2a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$

The normalization integral for the (1, 0, 0) wave function is

$$\int_0^\infty r^2 dr \int_0^\pi \sin\theta\, d\theta \int_0^{2\pi} d\phi \left|\psi_{1,0,0}(r,\theta,\phi)\right|^2 = \int_0^\infty 4a_0^{-3} e^{-2r/a_0} r^2 dr \int_0^\pi \tfrac{1}{2}\sin\theta\, d\theta \int_0^{2\pi} \tfrac{1}{2\pi} d\phi$$

$$= 4a_0^{-3} \int_0^\infty e^{-2r/a_0} r^2 dr = \frac{4}{a_0^3} \frac{2!}{(2/a_0)^3} = 1$$

The last integral is evaluated using the standard form $\int_0^\infty x^n e^{-ax} dx = n!/a^{n+1}$.

The normalization integral for the (2, 0, 0) wave function is

$$\int_0^\infty r^2 dr \int_0^\pi \sin\theta\, d\theta \int_0^{2\pi} d\phi \left|\psi_{2,0,0}(r,\theta,\phi)\right|^2 = \int_0^\infty \tfrac{1}{8} a_0^{-3} (2 - r/a_0)^2 e^{-r/a_0} r^2 dr \int_0^\pi \tfrac{1}{2}\sin\theta\, d\theta \int_0^{2\pi} \tfrac{1}{2\pi} d\phi$$

$$= \tfrac{1}{8} a_0^{-3} \int_0^\infty e^{-r/a_0} \left(4r^2 - 4\frac{r^3}{a_0} + \frac{r^4}{a_0^2}\right) dr = \tfrac{1}{8} a_0^{-3} \left[4\frac{2!}{(1/a_0)^3} - \frac{4}{a_0}\frac{3!}{(1/a_0)^4} + \frac{1}{a_0^2}\frac{4!}{(1/a_0)^5}\right] = \tfrac{1}{8}(8 - 24 + 24) = 1$$

Intrinsic Spin

7. Consider the energy level $n = 5$ of the hydrogen atom.
   a) Including the electron spin, what is the degeneracy of the n = 5 energy level of hydrogen?
   b) By adding up the number of states for each value of l permitted for n = 5, show that the same degeneracy as part (a) is obtained.

(a) degeneracy $= 2n^2 = 2(5)^2 = 50$

(b) For each value of l, the degeneracy is $2(2l+1)$.

$l = 0$: $2(0+1) = 2$
$l = 1$: $2(2+1) = 6$
$l = 2$: $2(4+1) = 10$
$l = 3$: $2(6+1) = 14$
$l = 4$: $2(8+1) = 18$
total: 50